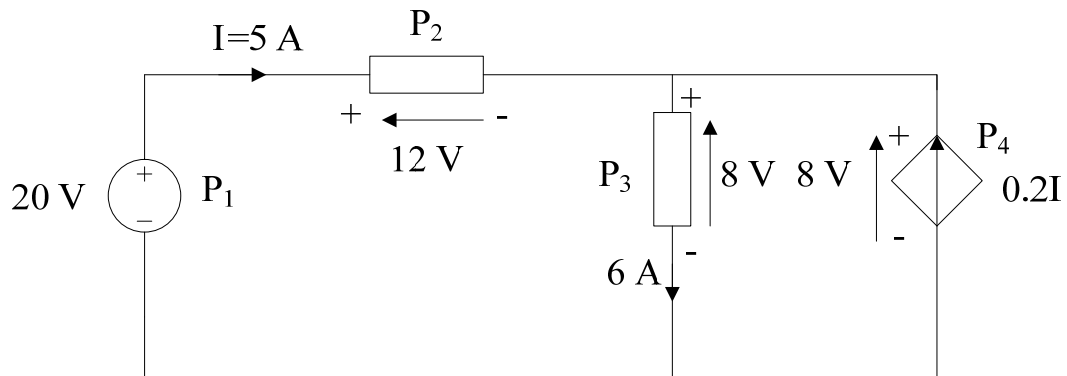


Calcolare le potenze che interessano ciascun bipolo, verificando che siano erogate o assorbite, in accordo con la convenzione dell'utilizzatore.



$$P_1 = 20(-5) = -100 \text{ W}$$

POTENZA EROGATA

$$P_2 = 12 \cdot 5 = 60 \text{ W}$$

POTENZA ASSORBITA

$$P_3 = 6 \cdot 8 = 48 \text{ W}$$

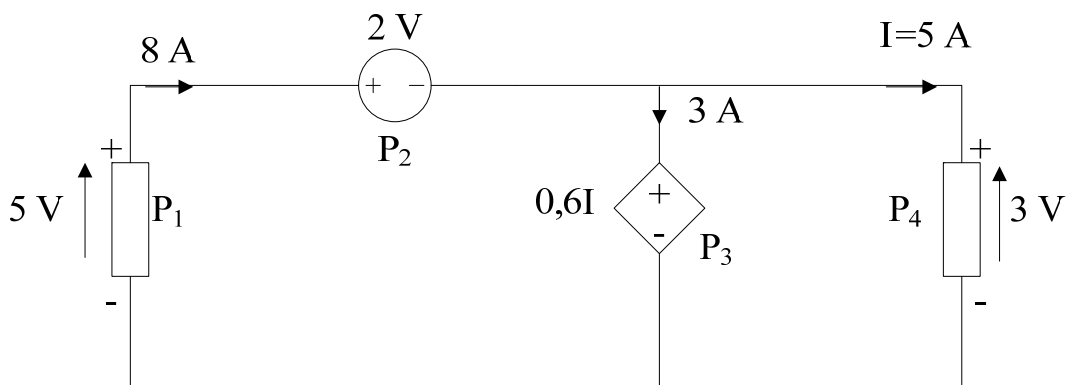
POTENZA ASSORBITA

$$P_4 = -8 \cdot 1 = -8 \text{ W}$$

POTENZA EROGATA

$$\sum P = 0 \rightarrow P_1 + P_2 + P_3 + P_4 = 0 \rightarrow -100 + 60 + 48 - 8 = 0$$

$$\sum P_a = \sum P_e \rightarrow P_2 + P_3 = P_1 + P_4 \rightarrow 60 + 48 = 100 + 8 = 0$$



$$P_1 = -8 \cdot 5 = -40 \text{ W}$$

POTENZA EROGATA

$$P_2 = 2 \cdot 8 = 16 \text{ W}$$

POTENZA ASSORBITA

$$P_3 = 3 \cdot 3 = 9 \text{ W}$$

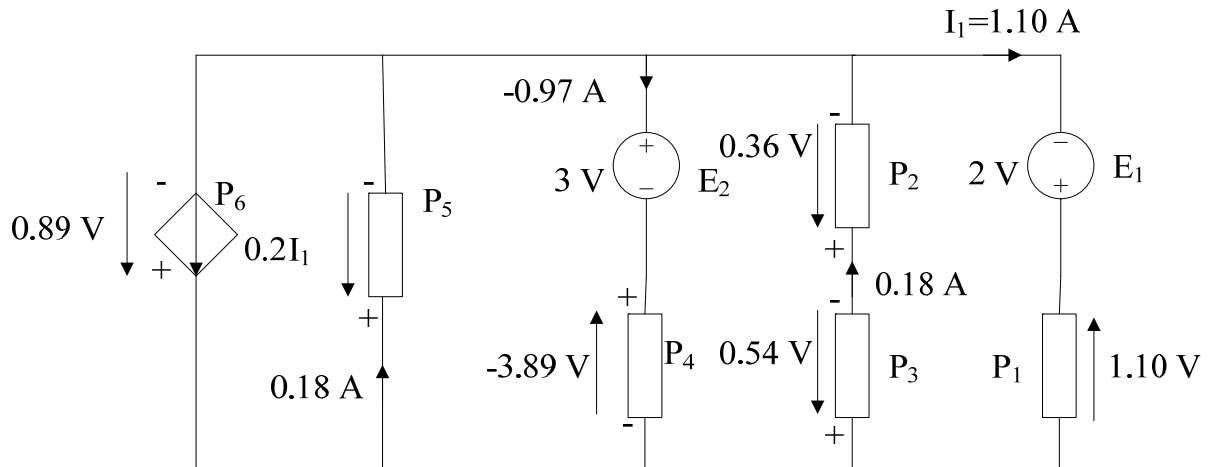
POTENZA ASSORBITA

$$P_4 = 3 \cdot 5 = 15 \text{ W}$$

POTENZA ASSORBITA

$$\sum P = 0 \rightarrow P_1 + P_2 + P_3 + P_4 = 0 \rightarrow -40 + 16 + 9 + 15 = 0$$

$$\sum P_a = \sum P_e \rightarrow P_2 + P_3 + P_4 = P_1 \rightarrow 16 + 9 + 15 = 40 = 0$$



$$P_6 = 0,89(-0,22) = -0,20 \text{ W}$$

POTENZA EROGATA

$$P_1 = 1,10 \cdot 1,10 = 1,21 \text{ W}$$

POTENZA ASSORBITA

$$P_{E1} = 2 \cdot (-1,10) = -2,20 \text{ W}$$

POTENZA EROGATA

$$P_2 = 0,36 \cdot 0,18 = 0,06 \text{ W}$$

POTENZA ASSORBITA

$$P_3 = 0,54 \cdot 0,18 = 0,10 \text{ W}$$

POTENZA ASSORBITA

$$P_{E2} = 3 \cdot (-0,97) = -2,91 \text{ W}$$

POTENZA EROGATA

$$P_4 = -3,89 \cdot (-0,97) = 3,77 \text{ W}$$

POTENZA ASSORBITA

$$P_3 = 0,18V_5$$

$$\sum P = 0 \rightarrow P_6 + P_1 + P_{E1} + P_2 + P_3 + P_{E2} + P_4 + P_5 = 0 \rightarrow -0,20 + 1,21 - 2,20 + 0,06 + 0,10 - 2,91 + 3,77 + P_5 = 0$$

$$P_5 = 0,17 \text{ W}$$

$$\sum P_a = \sum P_e \rightarrow P_1 + P_2 + P_3 + P_4 + P_5 = P_6 + P_{E1} + P_{E2} \rightarrow 1,21 + 0,06 + 0,10 + 3,77 + P_5 = 0,20 + 2,20 + 2,91$$

$$P_5 = 0,17 \text{ W}$$

$$P_5 = V_5 \cdot (0,18) \rightarrow V_5 = \frac{P_5}{0,18} = 0,94 \text{ V}$$